

Problem

The charge entering the positive terminal of an element is

$$q = 5 \sin 4\pi t \text{ mC}$$

while the voltage across the element (plus to minus) is

$$v = 3 \cos 4\pi t \text{ V}$$

(a) Find the power delivered to the element at $t = 0.3 \text{ s}$.

(b) Calculate the energy delivered to the element between 0 and 0.6 s.

Step-by-step solution

Step 1 of 2

The charge entering the positive terminal of an element, $q = 5 \sin 4\pi t \text{ mC}$.

The voltage across the element (plus to minus), $v = 3 \cos 4\pi t \text{ V}$.

(a)

Calculate the current.

$$\begin{aligned} i &= \frac{dq}{dt} \\ &= \frac{d}{dt}(5 \sin 4\pi t \text{ mC}) \\ &= (5)(4\pi) \cos 4\pi t \\ &= 20\pi \cos 4\pi t \text{ mA} \end{aligned}$$

The power delivered to the element at $t = 0.3 \text{ s}$ is,

$$\begin{aligned} p &= vi \\ &= (3 \cos 4\pi t \text{ V})(20\pi \cos 4\pi t \text{ mA}) \\ &= 60\pi \cos^2 4\pi(0.3) \end{aligned}$$

Use identity, $\cos^2(x) = \frac{1 + \cos 2(x)}{2}$ to simplify the equation further.

$$\begin{aligned} p &= 60\pi \left(\frac{1 + \cos 2(4\pi(0.3))}{2} \right) \\ &= \frac{60\pi}{2} + \frac{60\pi \cos(8\pi(0.3))}{2} \\ &= 94.24 + 29.124 \\ &= 123.36 \text{ mW} \end{aligned}$$

Therefore, the power delivered to the element at $t = 0.3 \text{ s}$ is 123.37 mW.

Step 2 of 2

(b)

The energy delivered to the element between 0 and 0.6 s.

$$\begin{aligned}w &= \int_0^{0.6} p \, dt \\&= \int_0^{0.6} 60\pi \cos^2 4\pi t \, dt \text{ mJ} \\&= 60\pi \int_0^{0.6} \left(\frac{1 + \cos 8\pi t}{2} \right) dt \text{ mJ}\end{aligned}$$

Simplify further.

$$\begin{aligned}w &= \frac{60\pi}{2} \left[t + \frac{1}{8\pi} \sin 8\pi t \right]_0^{0.6} \\&= \frac{60\pi}{2} \left[0.6 + \frac{1}{8\pi} (0.5877) \right] \\&= 56.54 + 2.204 \\&= 58.75 \text{ mJ}\end{aligned}$$

Therefore, the energy delivered to the element between 0 and 0.6s is 58.75 mJ .